

The Poset of Linear Subspaces of $\mathbb{R}^k$

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Transcribed from two-page handwritten notes. [Download the printable transcript \(PDF\)](#). The diagrams below are described in words as well as formulas.

Page 1: The poset and its realization

Let R^k denote the poset of proper, nontrivial linear subspaces of $\mathbb{R}^k$.

Examples. For $k = 2$,

$$|R^2| = \mathbb{RP}^1 \simeq S^1.$$

For $k = 3$, the realization is obtained from the space of flags between $G_1(\mathbb{R}^3)$ and $G_2(\mathbb{R}^3)$. The diagram is the pushout square with the flag space at the upper left, $G_1(\mathbb{R}^3)$ at the upper right, $G_2(\mathbb{R}^3)$ at the lower left, and $|R^3|$ at the lower right. The two maps from the flag space forget one of the two subspaces.

Goal.

$$|R^k| = S^{k(k+1)/2-2}.$$

Why look at this? The notes record

$$\Sigma^\infty C(k, V) \simeq \operatorname{holim}_{\operatorname{poset}} \bigvee_m S,$$

with $C(k, V)$ labeled “configuration space.” This can be analyzed using orthogonal calculus:

$$\Sigma_+^\infty \operatorname{Mor}(\mathbb{R}^n, V) \simeq \operatorname{holim}_{0 \neq E \leq \mathbb{R}^n} (\operatorname{hom}(\mathbb{R}^n, V) - \operatorname{hom}((\mathbb{R}^n, E), (V, 0))).$$

Here Mor is labeled “injective.” The next expression is

$$\simeq \operatorname{Nat}(|R^n| \downarrow -, F),$$

and the general formula alongside it is

$$\operatorname{holim}_{\mathcal{C}} F \simeq \operatorname{Nat}(|\mathcal{C}| \downarrow -, F(-)).$$

R^k is a category internal in $\mathbf{Top}$:

$$\text{space of objects} = \prod_{i=1}^{k-1} \operatorname{Gr}(i),$$

$$\text{space of morphisms} = \prod_{1 \leq i \leq j \leq k-1} \operatorname{Flag}(i; j).$$

The category diagram has source and target maps $s, t : \text{Mor} \rightarrow \text{Obj}$, an identity map in the opposite direction, and a composition map

$$\text{Mor} \times_{\text{Obj}} \text{Mor} \rightarrow \text{Mor}.$$

Form the simplicial nerve $N_{\bullet} R^k$. Its space $N_{\ell} R^k$ is the disjoint union of flags of length $\ell + 1$, and

$$|N_{\bullet} R^k| =: |R^k|.$$

Let $\text{Symm}(k)$ denote the space of $k \times k$ real symmetric matrices. Its dimension is $k(k + 1)/2$.

Define

$$f : N_0 R^k \rightarrow \text{Symm}(k)$$

by sending E to the orthogonal projection onto E . With respect to $E \oplus E^{\perp}$, this is the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Extend f linearly to a map

$$|N_{\bullet} R^k| \rightarrow \text{Symm}(k).$$

Page 2: Identification with a sphere

For $p \in |R^k|$, suppose p is in the $(d - 1)$ -simplex corresponding to the flag

$$0 < E_1 < \cdots < E_d < \mathbb{R}^k,$$

with coordinates

$$0 < \tau_1 < \cdots < \tau_{d-1} < 1.$$

Let $\tau_0 = 0$ and $\tau_d = 1$. Let

$$F_i = E_i \cap E_{i-1}^{\perp}, \quad F_0 = E_d^{\perp}.$$

Then $f(p)$ is the map that is multiplication by τ_i on F_i .

Include $\mathbb{R}$ into $\text{Symm}(k)$ as the “constant diagonal” matrices. This acts on $\text{Symm}(k)$ by addition, giving $\text{Symm}(k)/\mathbb{R}$.

There is a nice representative for each equivalence class: 0 is the smallest eigenvalue. Since $0 \in \text{Symm}(k)/\mathbb{R}$ has only one eigenvalue, $f(p) \neq 0$, so this descends to a map

$$|R^k| \rightarrow ((\text{Symm}(k)/\mathbb{R}) - \{0\})/\mathbb{R}_{>0} \simeq S^{k(k+1)/2-2}.$$

Representatives of classes have all eigenvalues with least 0 and greatest 1.

Then it is visibly a bijection. The target is Hausdorff and the source compact, so it is a homeomorphism. For example,

$$|R^3| = S^4.$$

Remark. Given a semisimple Lie group, there is a unique maximal compact, the quotient by which is a negatively curved symmetric space. Looking at flags of flat things embedded therein gives essentially the same idea.